Corpus Linguistics and Quantitative Language Laws

Corpus linguistics investigates empirical language patterns through large-scale, machine-readable text collections.

Key Statistical Principles:

1. Zipf\'s Law:
George Zipf observed that the frequency of any word is inversely proportional to its rank in the frequency table:
f(r) ~ 1 / r^s
A small cohort of common function words constitutes the overwhelming majority of total word tokens, while the vast majority of words appear with vanishing frequency (hapax legomena).

2. Heaps\' Law:
Heaps\' Law governs vocabulary growth as corpus size increases:
V(N) = k * N^beta
where V is vocabulary size, N is the number of tokens, and beta is typically between 0.4 and 0.6. This empirical law implies that vocabulary continuously expands sublinearly and never strictly saturates.

3. Statistical Semantics and Collocation:
John Rupert Firth famously stated: 'You shall know a word by the company it keeps.' Collocations quantify statistical co-occurrence preferences between lexical units using Pointwise Mutual Information (PMI) and Log-Likelihood Ratio tests.

These quantitative principles underpin modern corpus indexing, inverse document frequency thresholds, and statistical language modeling.